



**NOAA
FISHERIES**

2026 Sablefish CIE AFSC Longline Survey



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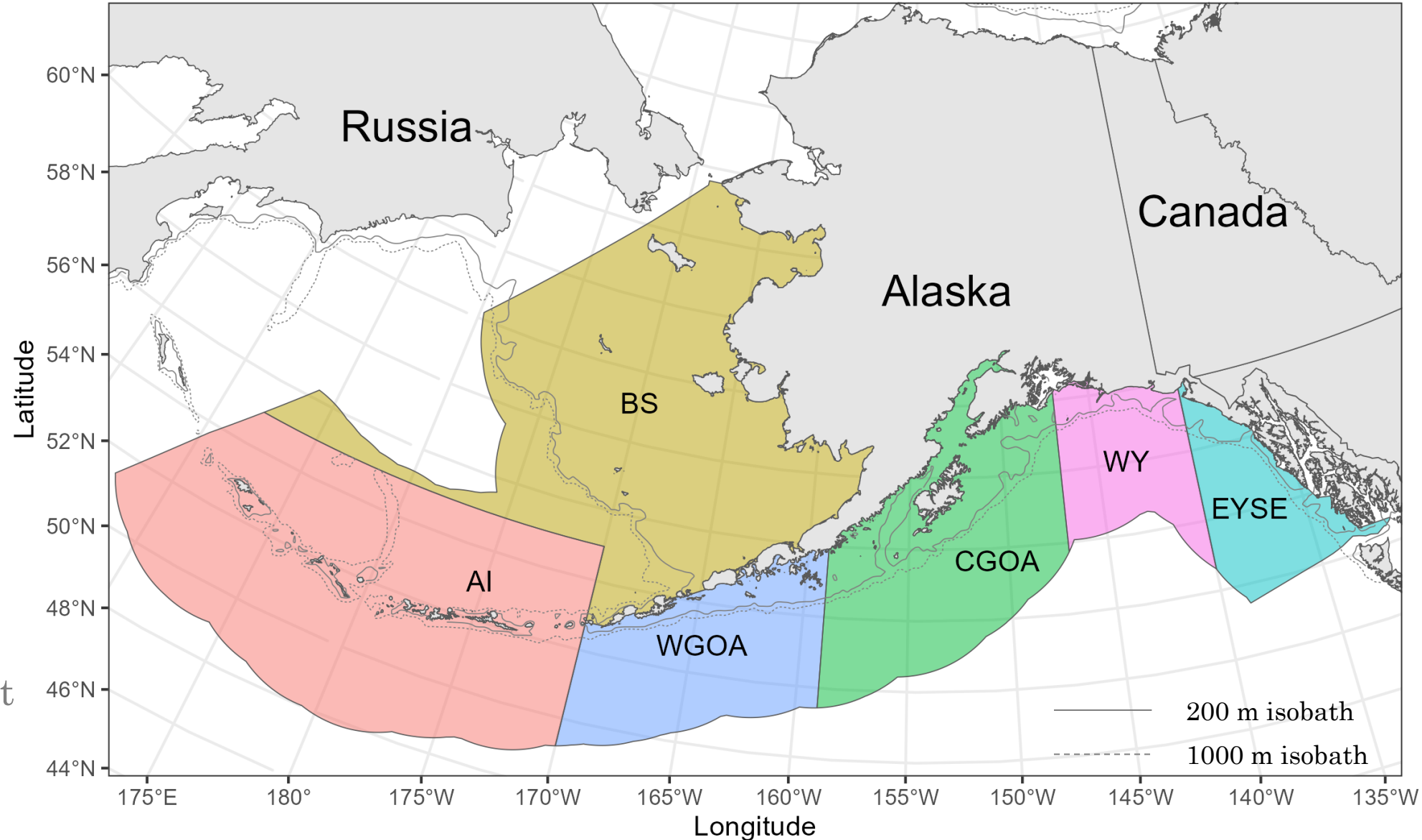


Outline

- Orientation and brief history of the survey
- Market-induced design change
- Sampling protocols
- Index calculations
- Whale depredation
- Handling regions not sampled
 - Gulf of Alaska: biennial starting in 2025
 - Bering Sea: biennial from 1997
 - Aleutian Islands: biennial from 1996
 - Western Aleutian Islands: last sampled in 1994

Management Areas

- BS = Bering Sea
- AI = Aleutian Islands
- WGOA = Western Gulf of Alaska
- CGOA = Central Gulf of Alaska
- WY = West Yakutat
- EYSE = East Yakutat/Southeast Alaska



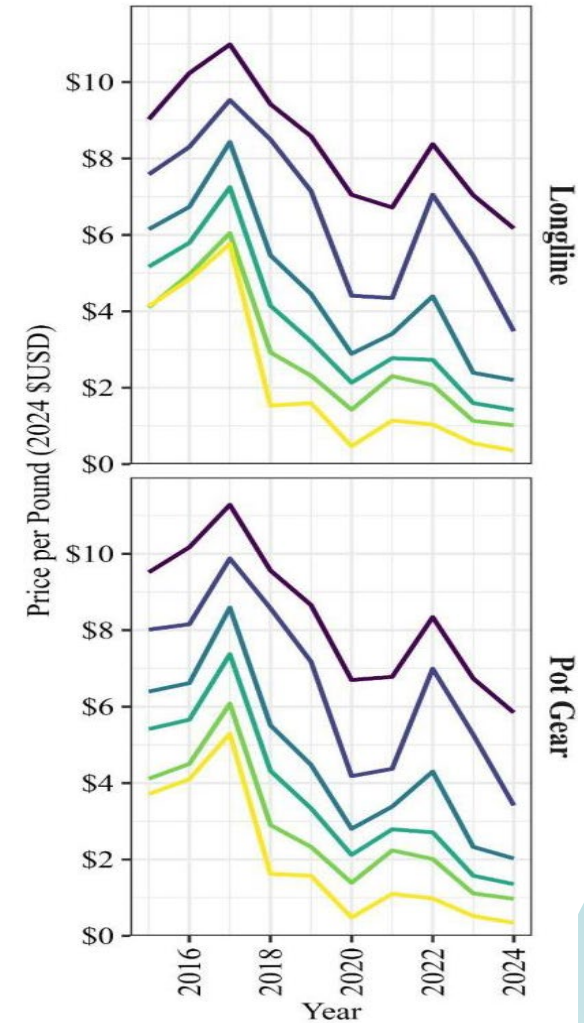
Longline Survey History

- Hook-and-line survey
 - Japan/US cooperative: 1979 to 1994
 - US: 1989 to present
 - GOA annual until 2023, now only odd years
 - BS/AI annual during coop survey, alternated biennially in 1996, and now both in even years
- Contracted freezer longliner samples Bering Sea, Aleutian Islands, and Gulf of Alaska continental slope 200 - 1,000m
- Before 2020 there were two vessel companies, but only one since
- Operated solely on a cost-recovery basis with sale of the catch covering vessel costs (to 2023)

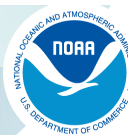


2024 Longline Survey Cancellation

- Economic value of sablefish declined due to record high abundance and poor markets
- Declining margins and losses in recent years
- Survey operated at an economic loss in 2023; projections for 2024 worse
- Government contract structure was unable to provide financial guarantee
- In April 2024, the survey was cancelled due to financial risk to contractor



Size Cat. ■ <2 ■ 2-3 ■ 3-4 ■ 4-5 ■ 5-7 ■ 7+



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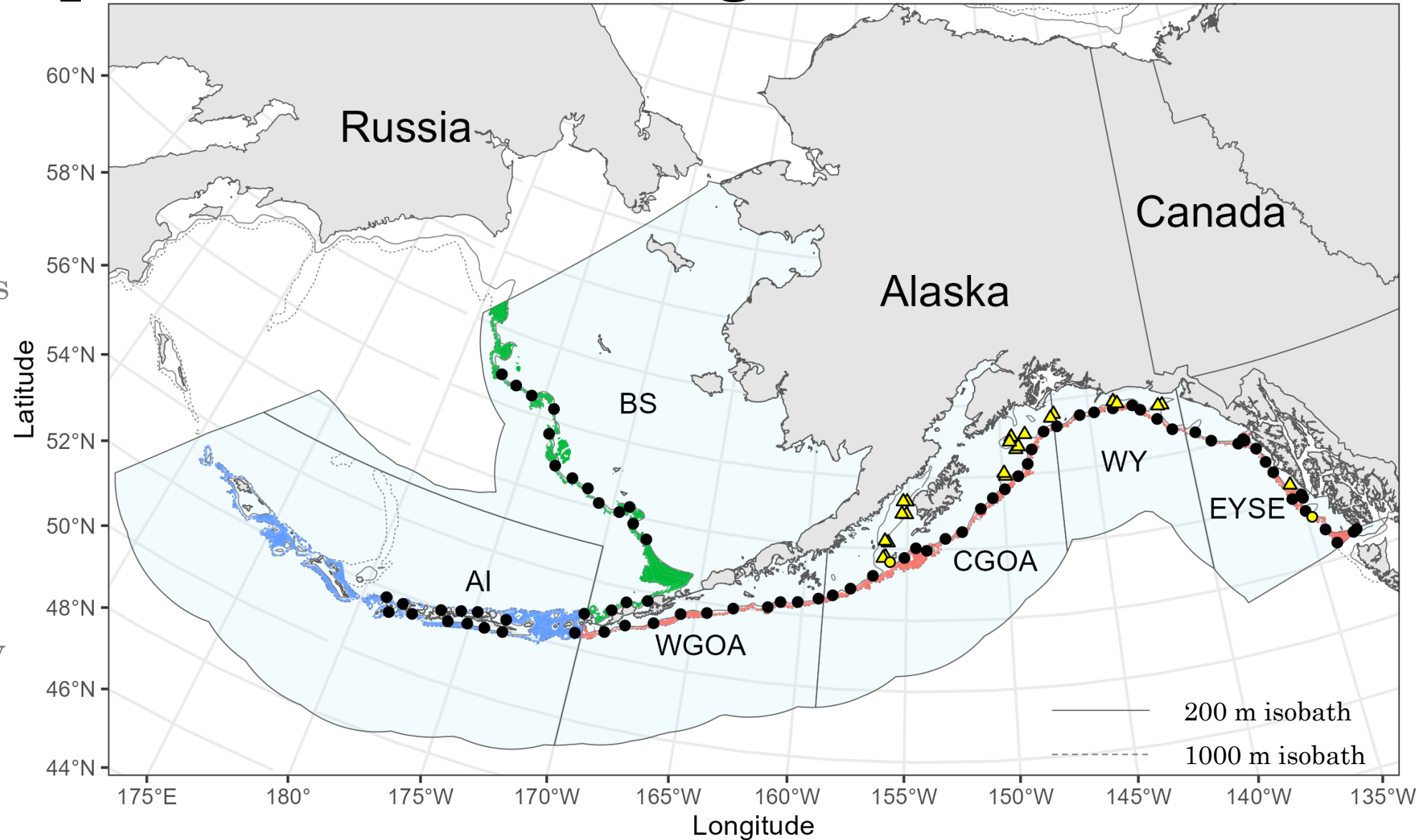
2025/26 Longline Survey Contract



- Restructured to guarantee a daily rate to remove market volatility risk to vessel
- Daily rate is offset by sale of retained catch to reduce net cost to government
- Few vessel-days available due to quota consolidation
- Survey days reduced and design re-structured
 - Previously sampled GOA every year and BS or AI every other year ~90 DAS
 - Now sample GOA in odd years for 50 DAS (2025) and BSAI in even years for 34 DAS (2026)

Survey footprint recent changes

- Red GOA footprint sampled odd years
- Green (BS) and blue (AI) footprint sampled even years
- Black circles are current stations
- Yellow circles are index stations lost in 2025
- Yellow triangles are non-index gully stations lost in 2025



Survey Protocols

One skate of gear:

- 45 hooks hand-baited with squid mantle
- 100-m length, 2-m spacing, 7-lb lead ball on ends

One set:

- 90 skates set shallow to deep
- Buoys and anchors at ends

One station:

- Standard slope station is 2 sets end to end
- Deep-gully is 1 set (6 stations in EGOA)
- Locations based on cooperative survey and spaced 30 to 60 km apart



Depth Strata	Min depth (m)	Max depth (m)
3	201	300
4	301	400
5	401	600
6	601	800
7	801	1000

Survey Protocols



Catch:

- 100% hook accounting: Species, baited/unbaited, ineffective (bent, broken, etc.), and associated depth.
- >5 hooks ineffective; skate not included in index calculations

Lengths:

- Random sample, depth-stratified, sex, 40-90K
- Other species include thornyheads, cod, and rockfish

Ageing:

- Random, if a sablefish is on first hook of a skate ~2.2%
- Depth-stratified, length, weight, sex
- Up to 1200 aged

Tagging:

- Random, skates # 10, 30, 50, 70, 100, 120, 140, and 160. ~4.4%, ~3,500
- Must be healthy with no injury, and record length

Whale depredation

- Determined by chief scientist with input from skipper and biologists as well as an assessment of damaged fish and catch rates before and after whales are present.
- When killer whale depredation is determined, the skate at which it starts through when it ends (usually end of the day) are removed before index calculations as sablefish catches can drop dramatically.
- When sperm whale depredation is determined, a depredation correction factor (1.18) is applied to the station so that estimates are not artificially deflated.



Relative Population Numbers (RPN)

Filter out skates with > 5 ineffective hooks or killer whale depredation

$\hat{\theta}_{i,j,k}$ = CPUE, mean number of fish per 45-hook skate by:

i = station

j = depth strata

k = geographic area

$$\hat{\theta}_{i,j,k} = \frac{n_{fish_{i,j,k}}}{n_{skate_{i,j,k}}},$$

$\bar{\theta}_{j,k}$ = mean CPUE by j and k

$n_{j,k}$ = number of station-level CPUE observations within j and k

$$E[\bar{\theta}_{j,k}] = \frac{\sum_i \hat{\theta}_{i,j,k}}{n_{j,k}},$$

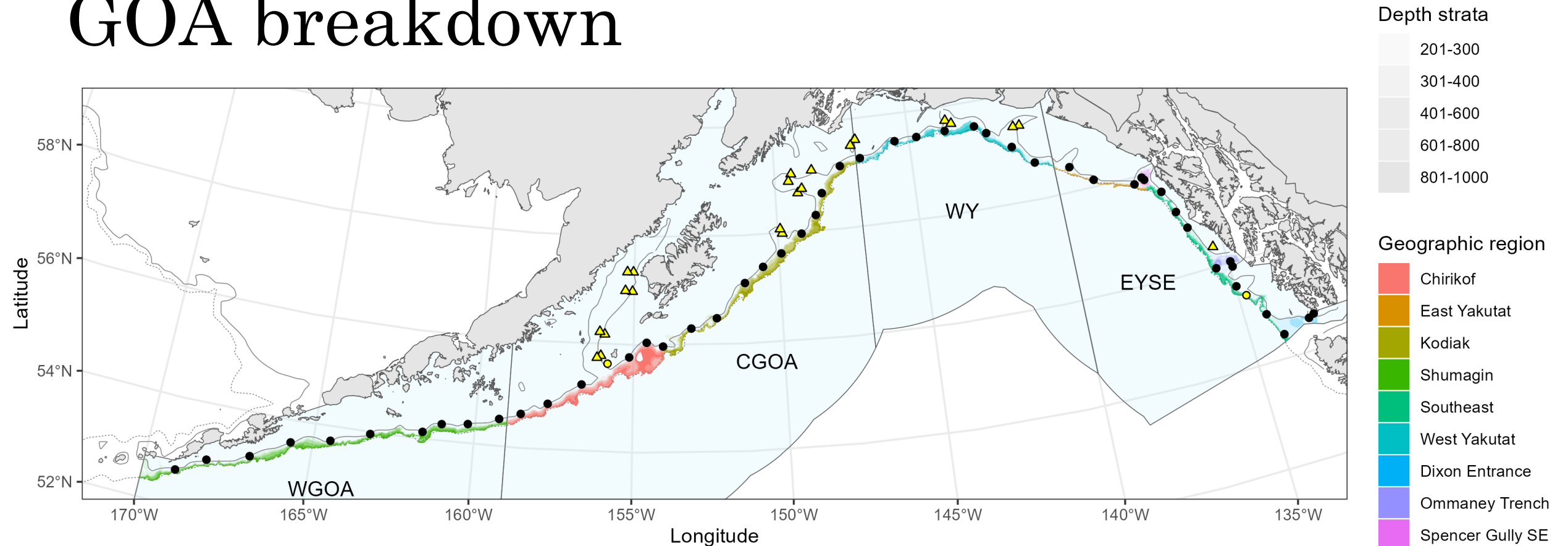
$\bar{\psi}_{j,k}$ = mean RPN by j and k

$w_{j,k}$ = area size (km²) by j and k

$$E[\bar{\psi}_{j,k}] = \bar{\theta}_{j,k} w_{j,k}$$

Can sum $\bar{\psi}_{j,k}$ to a geographic area, management area, FMP, Alaska-wide

GOA breakdown



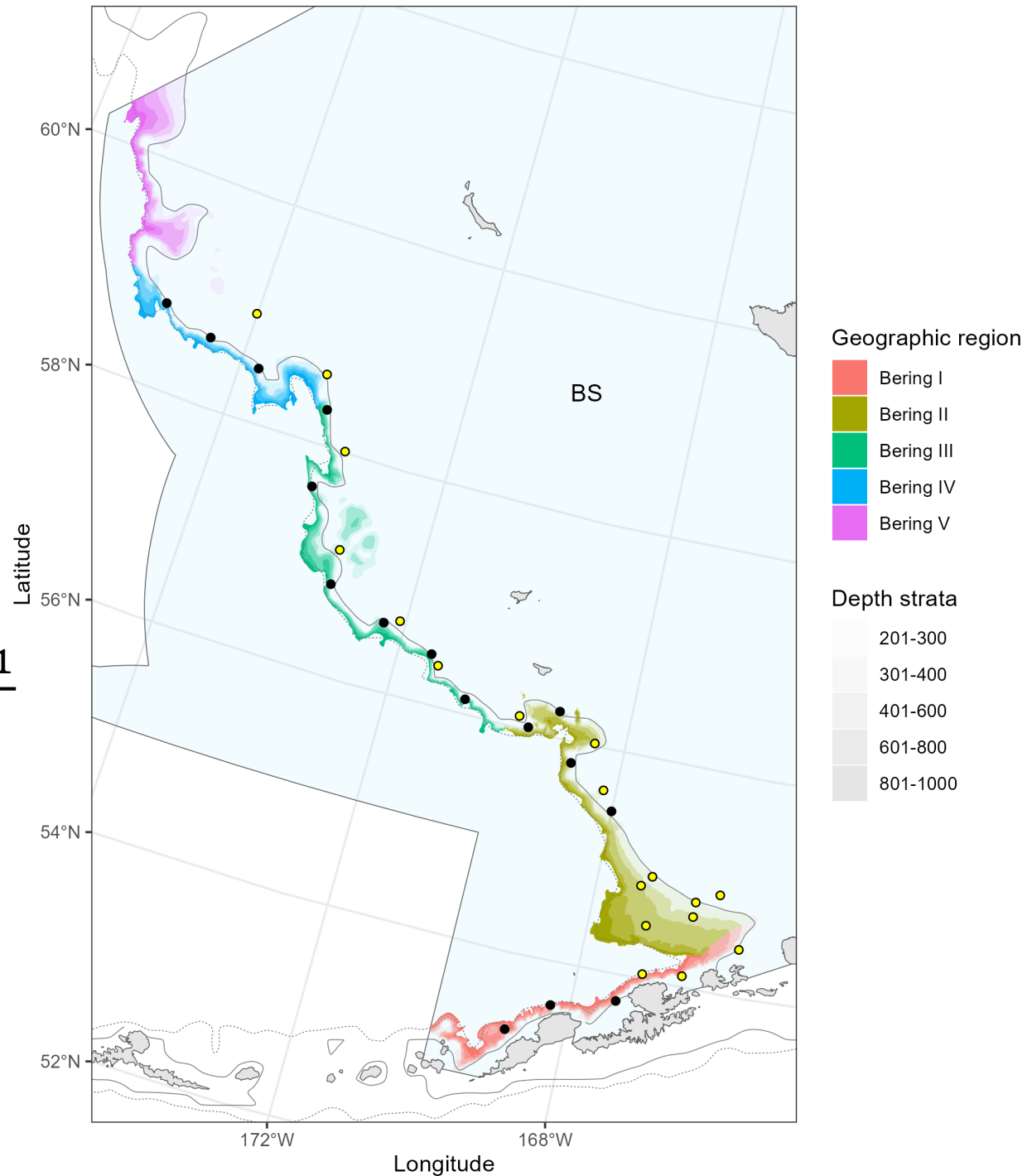
- Historically sampled annually, so always had an estimate
- Yellow circles are index stations lost in 2025
- Yellow triangles are non-index gully stations lost in 2025
- Annual change was used to extrapolate for off years BS and AI
- Now only sampled in odd years, so no extrapolation possible

BS breakdown

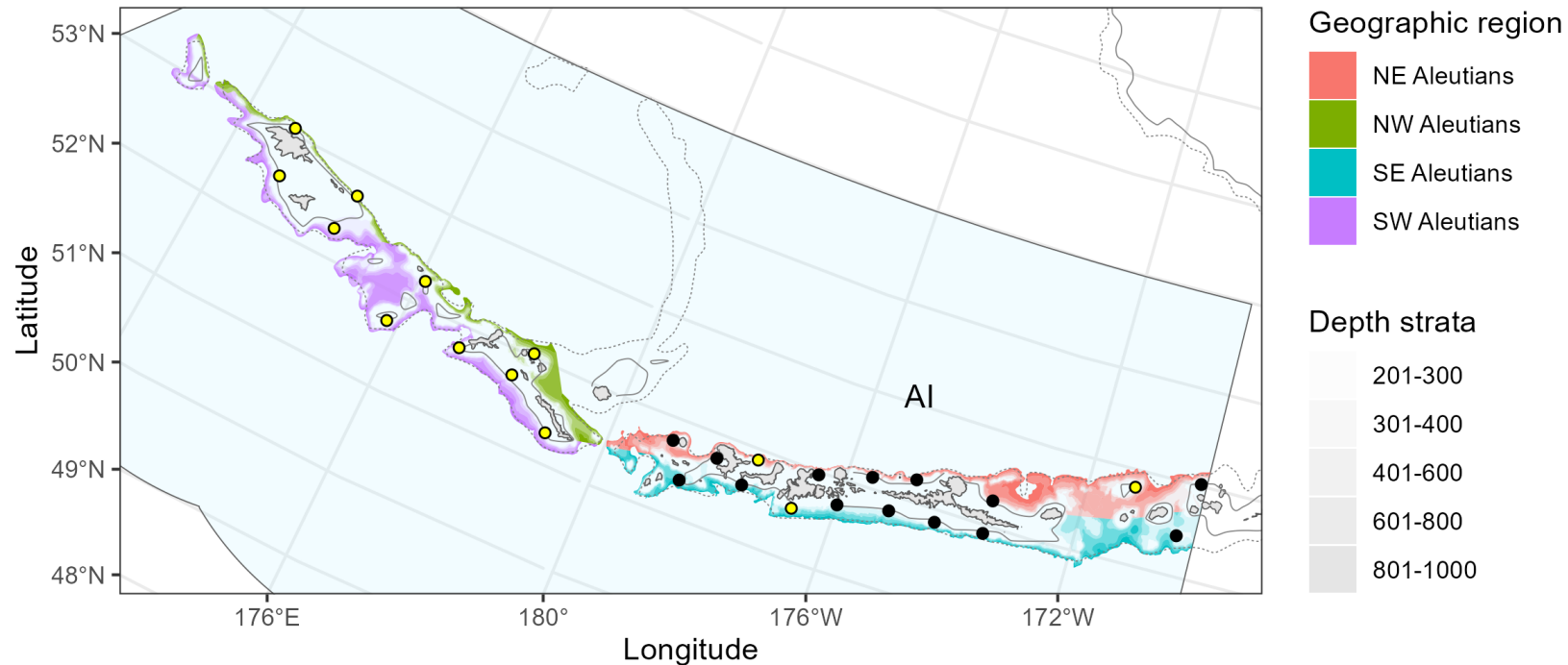
- Yellow circles were sampled by the cooperative survey before 1995.
- Black circles in odd years 1997 to 2023, and even year estimated from GOA change:

$$RPN_{BS,y} = RPN_{BS,y-1} \frac{RPN_{GOA,y} - RPN_{GOA,y-1}}{RPN_{GOA,y-1}}$$

- Now sampled in even years, and there is no annual GOA to do the old extrapolation



AI breakdown



- Yellow circles were sampled by the cooperative survey from 1983 to 1994, and no sampling since.
- Sampled in even years 1996 to 2022
- Now in even years

- Odd year estimated from GOA change:
$$RPN_{AI,y} = RPN_{AI,y-1} \frac{RPN_{GOA,y} - RPN_{GOA,y-1}}{RPN_{GOA,y-1}}$$
- Have extrapolated to WAI from EAI using fixed ratios of NWAI:NEAI = 0.1820 and SWAI:SEAI = 0.6581:
$$RPN_{AI,y} = 1.1820 * RPN_{NEAI,y} + 1.6581 * RPN_{SEAI,y}$$

Recommendations going forward

- Drop western AI extrapolation
 - We have not sampled the western AI in over 30 years, and there are numerous known issues in assuming an area-wide fixed ratio, we recommend dropping western AI extrapolation or pursue an updated method for filling in this missing area.
- Alaska-wide RPNs must be derived with alternative methods
 - We no longer have an annual survey in any region so we cannot use an annual change to extrapolate to off-years in unsampled regions. A carryover RPN from the last sampled year could be used, e.g.,

$$RPN_{AK,2026} = RPN_{AI,2026} + RPN_{BS,2026} + RPN_{GOA,2025}$$

- Work with assessment author to prioritize sampling needs and adjust to the new regional sampling frequencies and seek alternative methods for modelling this Alaska-wide stock.

Thank You!

2026 CIE reviewers
Vessels and crews
NOAA survey staff
NOAA assessment team

